

Yifan Dong

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Education

Purdue University, West Lafayette, IN

Aug 2023 – Present

Ph.D. in Electrical and Computer Engineering

- Research interest: Optimization and Machine Learning for Power Systems and Electricity Markets
- GPA: 3.84/4.0 [Transcript]
- Coursework: Computational Method in Optimization, Convex Optimization, Computational Method for Power System Analysis, Optimization for Deep Learning, Algorithm Design, Analysis and Implementation, etc.

North China Electric Power University, Beijing, China

Aug 2019 – June 2023

B.Eng. in Electrical Engineering and Automation

- GPA: 4.31/5.0 (Top 1%) [Transcript]
- Coursework: Power System Analysis, Power System Economy and Management, Automatic Control Theory, etc.

Experiences

Independent System Operator - New England, Holyoke, MA

June 2026 – Aug 2026

Student Intern at Advanced Technology Solutions (ATS)

Mentors: Dr. Feng Zhao, Dr. Xiaochu Wang, and Dr. Yufan Zhang

- *Project 1: Bidding Behavior in the Day-Ahead Ancillary Services (A/S) Market*
 - Conducted **theoretical and statistical analyses of A/S offers**, characterizing offer-price patterns, evaluating offer competitiveness, and examining relationships between offers and award outcomes.
 - Developed a **data-driven framework for unit classification** that learns low-dimensional representations of daily bidding behavior, identifies recurring A/S offer patterns through unsupervised clustering, and groups units by their recurring bidding profiles.
- *Project 2: In-House Probabilistic Evaluation Tool for Resource Adequacy (PETRA) Development*
 - Evaluated **input consistency between PETRA and GE MARS**, identifying and resolving conversion discrepancies to establish a reliable basis for subsequent output validation.
 - Validated **PETRA's resource-adequacy modeling and dispatch logic** using analytical benchmarks and production-scale test cases, examining loss-of-load behavior, energy-storage dispatch, load-shedding strategies, and multi-scenario reliability outcomes.

ASPIRE Research Center, Purdue University, West Lafayette, IN

Aug 2023 – Present

Research Assistant | Advisor: Prof. Junjie Qin

- *Topic 1: Privacy-Preserving Demand Flexibility Aggregation for Flexible Loads*
 - Designed a **privacy-preserving framework for aggregating demand flexibility**, enabling large-scale coordination among devices while keeping user data local and secure.
 - Re-architected a bilevel optimization problem into a **scalable single-level learning framework**, enabling efficient end-to-end optimization for large-scale flexibility aggregation.
 - Developed a **federated gradient-based algorithm** with backend logic for secure exchange of anonymized model parameters, reducing communication overhead and preserving data privacy.
 - Conducted extensive testing and performance evaluation, **demonstrating significant improvements in system efficiency and aggregate flexibility** over existing baselines.
- *Topic 2: Real-Time Charging Optimization System for Electric Roadways*
 - Designed and implemented a **real-time charging control system** for electric roadways, enabling safe, efficient wireless power transfer between roadway microgrids and electric vehicles under dynamic traffic conditions.
 - Re-architected the charging-control problem using **dynamic programming (DP)**, exposing **state-transition structure and constraints** that enabled a simplified and implementable solution strategy.
 - Developed a **causal, online scheduling algorithm** that exploits the DP-derived properties to support **lightweight, real-time decisions** without relying on future information.
 - Evaluated the system across diverse traffic scenarios, demonstrating **low suboptimality, reliable constraint satisfaction, and strong robustness to prediction errors**.

Purdue ECE, West Lafayette, IN

Fall 2024, Spring 2026

Teaching Assistant for ECE 31032-Power Systems Engineering

- Supported instruction on **power flow, transmission lines, transformers, and economic dispatch.**
- Held weekly office hours, designed assignments and exams, and graded projects.

Publications

- [1] **Y. Dong**, G. Chen, J. Qin, "Federated Aggregation of Demand Flexibility", *IEEE Transactions on Smart Grid*, 2026.
- [2] **Y. Dong**, G. Chen, J. Qin, S. Sivaranjani, X. Lu, D. Aliprantis, D. Love, "Real-Time Charging Control for Electric Roadways", *IEEE Transactions on Smart Grid*, 2026.
- [3] **Y. Dong**, J. Qin, S. Sivaranjani, X. Lu, D. Aliprantis, D. Love, "Real-Time Charging Control for Electric Roadways: Formulation and Causal Algorithms", *IEEE Power & Energy Society General Meeting*, 2024.

Presentations

- [1] "Federated Aggregation of Demand Flexibility", 2025 ASPIRE Annual Meeting, Logan, UT [Poster]
- [2] "Real-Time Charging Control for Electric Roadways", 2024 ASPIRE Annual Meeting, Logan, UT [Poster]
- [3] "Real-Time Charging Control for Electric Roadways", 2024 IEEE PES General Meeting, Seattle, WA [Poster]

Technical Skills

Python, SQL, MATLAB, PyTorch, scikit-learn, Pandas, NumPy, CVXPY, Gurobi, CPLEX, SUMO